Trust confidence sets: statistical inference for selecting models under several trustworthiness criteria

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Abstract

Trustworthy machine learning is usually assessed with several criteria at once: predictive error, group fairness, calibration and robustness to perturbation. Candidate models are then compared through point estimates computed on one finite test sample, and the "winner" or the Pareto front of those estimates is reported as if it were a population fact. We argue that this practice has no statistical guarantee and propose the *trust confidence set* (TCS): a data-dependent set of models that contains every population-nondominated model with probability at least $1 - \alpha$, asymptotically. The construction uses paired bootstrap resampling of the test instances, studentised pairwise differences, an intersection–union step across criteria and a maximum over rival models. In a Monte Carlo study with eight models and three criteria, the TCS retained all population-nondominated models in at least 99.7% of replications (nominal level 95%), whereas the Pareto front of point estimates retained all of them in only 34–42% of replications. On two public benchmarks (Adult, $n_{\text{test}} = 19{,}537$; German credit, $n_{\text{test}} = 400$), no model could be eliminated when four criteria were considered jointly, although the Pareto front of point estimates excluded one model on Adult and two on German credit, and the identity of the best model under an equal-weight composite score changed across bootstrap resamples (on Adult the best model under an equal-weight composite score was one of four different models across resamples, with frequencies of 41%, 41%, 12% and 5%). We also report the bootstrap probability of Pareto-front membership as a simple stability index. The results show that trust-oriented model comparison needs an explicit uncertainty statement, and that for realistic test-set sizes most differences between competent models cannot be resolved.

Keywords: trustworthy machine learning; model selection; Pareto dominance; model confidence set; bootstrap; algorithmic fairness; calibration; robustness

1 Introduction

Predictive accuracy is no longer the only quantity of interest when a machine learning model is deployed. Regulators, auditors and practitioners ask whether the model is accurate, whether its errors are distributed fairly across groups, whether its probabilities are calibrated and whether its output is stable under perturbation of the input. The Special Issue on Statistics for Data Science and AI frames this agenda as making models "accurate, explainable, robust and fair", and asks for methods that measure trustworthiness in real applications. A growing literature

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supplies the measures: equality of opportunity (Hardt et al., 2016), expected calibration error (Guo et al., 2017), and rank-based families of metrics for accuracy, explainability, robustness and fairness (Giudici et al., 2024; Babaei et al., 2025; Wu, 2026).

What this literature does not supply is an inferential layer for the *comparison* of models on these measures. A trust measure computed on a test sample is an estimate. Two candidate models may differ by less than the sampling noise on one criterion, and may reverse order on another. When the selection rule is "smallest composite score" or "member of the Pareto front of the estimates", its output is a random set or a random index, and the report rarely says how variable that is. The problem is old in forecasting and econometrics, where the reality-check and superior-predictive-ability tests (White, 2000; Hansen, 2005) and the model confidence set (Hansen et al., 2011) replace a single winner by a set that is guaranteed to contain the best model with stated confidence. Those tools handle one loss function. Trustworthiness is intrinsically vector-valued.

Contribution. We define a population notion of nondominated models under a vector of trust criteria, and a *trust confidence set* (TCS) that contains all such models with asymptotic probability at least $1-\alpha$. The procedure needs only per-instance predictions on a common test sample and a paired bootstrap, so it applies to any model class and any criteria that can be written as functionals of the empirical distribution. We (i) prove asymptotic control under bootstrap consistency, (ii) show by simulation that the Pareto front of point estimates fails badly at the same task, and (iii) apply the method to two public benchmarks to show how many models survive. We also give a Pareto-membership probability as a descriptive companion.

2 Related work

Statistical comparison of learners. Significance testing for classifiers on one or several datasets is well established (Dietterich, 1998; Demšar, 2006), and Bayesian alternatives have been argued to be more informative than null-hypothesis tests (Benavoli et al., 2017). Bouthillier et al. (2021) show that benchmark conclusions depend on several sources of variance (data sampling, initialisation, hyper-parameter search), and that ignoring them inflates claimed improvements. These works compare models on one performance metric.

Confidence sets for model selection. Hansen, Lunde and Nason (2011) introduce the model confidence set, a set of models that contains the best model with a given probability and that grows when the data are uninformative. The tests of White (2000) and Hansen (2005), and the step-down multiple-testing framework of Romano and Wolf (2005), supply the bootstrap machinery for maxima of studentised statistics over many comparisons. Our construction borrows this machinery but targets nondominance in a vector of criteria rather than minimality of one loss.

Multi-objective model selection. Pareto-based selection is common in machine learning. SPRINT-Race (Zhang et al., 2018) is, to our knowledge, the closest statistical treatment: a racing algorithm that returns Pareto-optimal models with a fixed-confidence guarantee under a

sequential probability ratio test with an indifference zone. It is designed to save computation during a search, whereas the TCS is a post-hoc confidence set for a fixed list of candidates and uses the paired structure of a shared test sample. In a search of the literature available to us we did not find a bootstrap confidence set for Pareto-optimal models under trust criteria, but our search was not exhaustive.

Uncertainty in fairness and trust measures. Group-fairness estimates have large variance for small groups. Miller et al. (2021) propose model-based estimators and an approximate bootstrap for subpopulation metrics, and Ferrara et al. (2025) derive a central limit theorem for statistical parity difference with a Wald test and a Bayesian estimator for small intersectional groups. Those papers quantify the uncertainty of *one* fairness metric for *one* model. We use bootstrap standard errors for pairwise differences of arbitrary criteria across models.

Model multiplicity. Underspecification (D'Amour et al., 2022) and predictive multiplicity (Marx et al., 2020) document that many models with near-identical accuracy can differ in fairness, robustness and individual predictions. This is the setting in which a point-estimate winner is least reliable, and in which the TCS is meant to report which differences can and cannot be resolved with the available test data.

Trust metrics in this journal. Recent contributions to the *International Journal of Data Science and Analytics* include robustness measures for linear and nonlinear models (Giudici et al., 2024), fairness metrics for credit lending (Wu, 2026), and evaluation of imputation methods by downstream predictive performance (Nikitina and Mouriño, 2026). The Rank Graduation Box (Babaei et al., 2025) unifies accuracy, explainability, robustness and fairness through rank-based measures. Our method is complementary: it can take any of these measures as a coordinate of the criterion vector and says how much of a difference between models is resolved by the data.

3 Method

3.1 Setting and target

Let $\mathcal{M} = \{1, \dots, M\}$ be candidate models, already trained, and let $D = \{z_i\}_{i=1}^{n}$ be a test sample of independent instances. For each model m and criterion $k = 1, \dots, K$ let $\hat{c}_{mk} = \phi_k(\hat{F}_m)$ be an estimate of $\mu_{mk} = \phi_k(F_m)$, where F_m is the population law of the (instance, prediction, perturbed prediction, group) record produced by model m and ϕ_k is a smooth functional. All criteria are oriented so that smaller is better (error, equalised-odds gap, calibration error, performance loss under perturbation). Inference is conditional on the trained models; Section 6 discusses training variability.

Definition. Model m is *strictly dominated* by model j if $\mu_{mk} > \mu_{jk}$ for every k. The population nondominated set is $\mathcal{P}_0 = \{m : \text{ no } j \text{ strictly dominates } m\}$. This definition is deliberately the weakest notion of domination: a model tied with a rival on a single criterion belongs to $\mathcal{P}_0$. It is the notion for which an exclusion can be certified by a test, since declaring m worse requires evidence of difference on *every* criterion.

Table 1: Criteria used in the experiments. Each is a functional of the empirical distribution of one model's test records.

Criterion	Definition
Error	misclassification rate
EO gap	$\max\{\lvert\mathrm{TPR}_0 - \mathrm{TPR}_1\rvert, \lvert\mathrm{FPR}_0 - \mathrm{FPR}_1\rvert\}$ across the two groups (Hardt et al., 2016)
ECE	10-bin expected calibration error of the predicted probability (Guo et al., 2017)
Noise drop	error on the perturbed test sample minus error on the clean sample

3.2 Procedure

Let $d_{mjk} = \hat{c}_{mk} - \hat{c}_{jk}$ (positive means m is worse). Draw B paired bootstrap samples of the test instances, applying the same resample to all models, and recompute all estimates to obtain $\hat{c}^{*b}_{mk}$. With $d^{*b}_{mjk} = \hat{c}^{*b}_{mk} - \hat{c}^{*b}_{jk}$ and $\hat{s}_{mjk}$ the bootstrap standard deviation of d^{*b}_{mjk}:

1. the studentised difference is $t_{mjk} = d_{mjk}/\hat{s}_{mjk}$ and the centred bootstrap version is $t^{*b}_{mjk} = (d^{*b}_{mjk} - d_{mjk})/\hat{s}_{mjk}$;
2. the dominance statistic of model m is $S_m = \max_{j \neq m} \min_k t_{mjk}$ (an intersection–union statistic across criteria, a maximum over rivals);
3. the critical value $\hat{q}_{1-\alpha}$ is the $(1-\alpha)$-quantile over b of $\max_{m \neq j,\, k} t^{*b}_{mjk}$;
4. $\mathrm{TCS}_{1-\alpha} = \{m : S_m \leq \hat{q}_{1-\alpha}\}$.

A pair (m, j) with $\hat{s}_{mjk} = 0$ (identical predictions on criterion k) is assigned $t_{mjk} = 0$.

Proposition 1. *Suppose that, jointly over (m, j, k), the bootstrap law of $\hat{s}^{-1}_{mjk}(d^{*}_{mjk} - d_{mjk})$ is consistent for the law of $\hat{s}^{-1}_{mjk}(d_{mjk} - \delta_{mjk})$ with $\delta_{mjk} = \mu_{mk} - \mu_{jk}$, and that the limit has a continuous distribution function at its $(1-\alpha)$-quantile. Then $\limsup_{n\to\infty} \mathbb{P}(\mathcal{P}_0 \not\subseteq \mathrm{TCS}_{1-\alpha}) \leq \alpha$.*

Sketch. Fix $m \in \mathcal{P}_0$. For every rival j some criterion $k(j)$ satisfies $\delta_{mjk(j)} \leq 0$, so $t_{mjk(j)} \leq z_{mjk(j)} := \hat{s}^{-1}_{mjk(j)}(d_{mjk(j)} - \delta_{mjk(j)})$. Hence $S_m \leq \max_j t_{mjk(j)} \leq \max_{j,k} z_{mjk} \leq Z := \max_{m',j,k} z_{m'jk}$. The model m is excluded only if $S_m > \hat{q}_{1-\alpha}$, which requires $Z > \hat{q}_{1-\alpha}$. By the consistency assumption, $\hat{q}_{1-\alpha}$ converges to the $(1-\alpha)$-quantile of the limit law of Z, so $\mathbb{P}(Z > \hat{q}_{1-\alpha}) \to \alpha$. A union over $m \in \mathcal{P}_0$ is not needed because the bound on Z is common to all models. □

Remark 1 (Conservativeness)**.** *The critical value takes a maximum over criteria rather than a minimum because a model in $\mathcal{P}_0$ may be tied with a rival on a single criterion while clearly worse on the others (the least favourable configuration of an intersection–union test). Using the minimum under a global null that sets all differences to zero gives a smaller critical value; in a preliminary run of the simulation design of Section 4, that variant retained all of $\mathcal{P}_0$ in only 85–94% of replications when $\rho = 0$, below the nominal level, which is why it is not used. The price is lower power; sharper step-down variants (Romano and Wolf, 2005) are a natural extension.*

Remark 2 (Companion indices)**.** *Two descriptive quantities accompany the set. The Pareto-membership probability $\hat{\pi}_m = B^{-1} \sum_b \mathbf{1}\{m \text{ is on the Pareto front of } \hat{c}^{*b}\}$, and, for any scalarisation (here an equal-weight sum of criteria divided by their across-model standard deviation), the bootstrap distribution of the rank and of the arg-min. These carry no coverage guarantee; they show how unstable a point-estimate decision is.*

4 Simulation

Design. There are $M = 8$ models and $K = 3$ criteria. Population means (smaller is better) for models 0–3 are $(0.10, 0.10, 0.30)$, $(0.30, 0.10, 0.10)$, $(0.10, 0.30, 0.10)$ and $(0.20, 0.20, 0.20)$; with $\mu_3 = (0.20, 0.20, 0.20)$ and margin $\delta \in \{0.02, 0.05\}$, model 4 is $\mu_3 + \delta(1, 1, 1)$ and model 5 is $\mu_3 + 2\delta(1, 1, 1)$ (both strictly dominated by model 3), model 6 is $\mu_3 + \delta(1, 1, 0)$ (tied with model 3 on criterion 3, hence in $\mathcal{P}_0$) and model 7 is $\mu_3 + \delta(1, 1, -1)$ (a trade-off, in $\mathcal{P}_0$). Per-instance values are Gaussian with standard deviation 0.15, correlation $\rho \in \{0, 0.5\}$ between criteria within a model and 0.5 between models. Test sizes are $n \in \{200, 500, 2000\}$, with 400 replications per cell, $B = 300$ bootstrap samples and $\alpha = 0.05$. The naive competitor is the Pareto front of the point estimates. Two outcomes are recorded: whether *all* models of $\mathcal{P}_0$ are retained, and the share of strictly dominated models that are excluded.

Results. Table 2 and Figure 1 give the results. The TCS retained all models of $\mathcal{P}_0$ in at least 99.75% of replications in every cell; the sampling error of an estimated proportion with 400 replications is about one percentage point, and the proportions are well above the nominal 95%, which is consistent with the conservativeness discussed above. The naive Pareto front retained all of $\mathcal{P}_0$ in 33.5–42.3% of replications, and the proportion did not improve with n: the tied model 6 is on the estimated front only when sampling noise happens to favour it on criterion 3. The naive front excluded dominated models more often than the TCS at small n (for example $n = 200, \delta = 0.02$: 0.97 versus 0.15–0.23), which is exactly the behaviour of an unfounded decision rule. The TCS exclusion share rises with n and δ and reaches one at $n = 2000$ (0.995 or higher in all cells).

Table 2: Monte Carlo results (400 replications per row). Retention: probability that all population-nondominated models are retained (nominal 0.95). Exclusion: share of strictly dominated models removed. TCS size: mean number of models retained (population nondominated: 6).

n	δ	ρ	Retention		Exclusion		TCS size
			TCS	Naive	TCS	Naive	
200	0.02	0.0	1.000	0.422	0.15	0.97	7.69
200	0.02	0.5	0.998	0.362	0.23	0.98	7.53
200	0.05	0.0	1.000	0.385	0.87	1.00	6.25
200	0.05	0.5	0.998	0.410	0.92	1.00	6.15
500	0.02	0.0	1.000	0.355	0.51	1.00	6.97
500	0.02	0.5	1.000	0.335	0.57	1.00	6.86
500	0.05	0.0	0.998	0.380	1.00	1.00	6.00
500	0.05	0.5	1.000	0.352	1.00	1.00	6.00
2000	0.02	0.0	1.000	0.352	0.99	1.00	6.01
2000	0.02	0.5	1.000	0.388	1.00	1.00	6.01
2000	0.05	0.0	1.000	0.362	1.00	1.00	6.00
2000	0.05	0.5	1.000	0.355	1.00	1.00	6.00

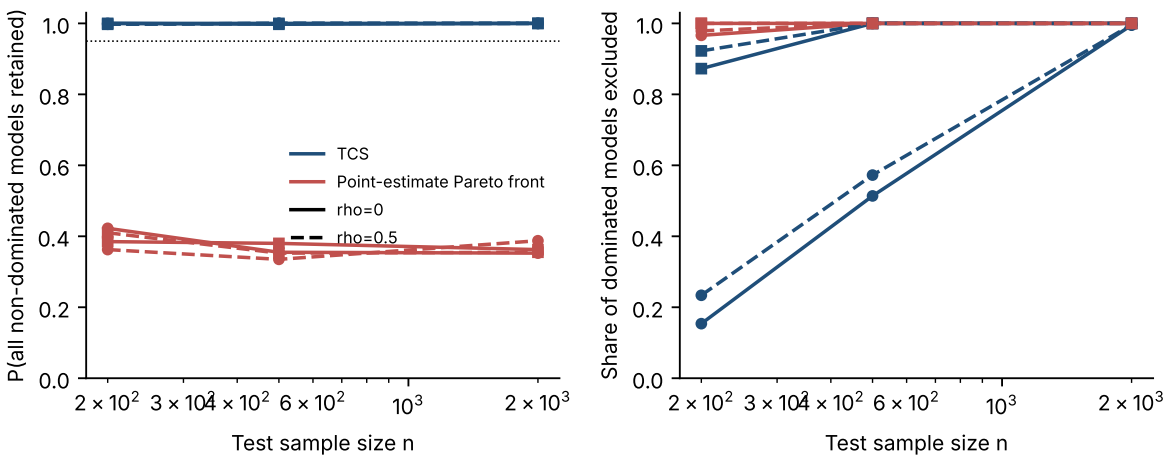


Figure 1: Retention (left) and exclusion (right) as a function of test size. Blue: TCS. Red: Pareto front of point estimates. Solid: $\rho = 0$; dashed: $\rho = 0.5$; circles: $\delta = 0.02$; squares: $\delta = 0.05$.

5 Case studies

Data and models. We use the Adult census income data (OpenML, version 2; 48,842 records; sensitive attribute: sex, removed from the features) and the German credit data (OpenML, version 1; 1,000 records; group: age below 26, a frequently used convention, with age removed from the features). Each data set is split once with a stratified 60/40 train/test split (seed 0), giving $n_{\text{test}} = 19{,}537$ and 400. Nine models are trained with scikit-learn defaults unless stated: logistic regression, logistic regression with $C = 0.01$, decision trees of depth 3 and 12, random forest (200 trees, minimum leaf 3), histogram gradient boosting, 25-nearest neighbours, Gaussian naive Bayes and a two-layer perceptron (64, 32 units, early stopping). Numeric features are standardised and categorical features one-hot encoded. For the noise criterion, numeric test features are perturbed once with Gaussian noise of standard deviation equal to half the training standard deviation. We use $B = 1000$ bootstrap samples and $\alpha = 0.05$. The fairness measures are used only as an illustration of vector-valued comparison and say nothing about the fitness of these models for decisions about individuals.

Results. Tables 3 and 4 and Figure 2 list the point estimates, the dominance statistics S_m against the critical values ($\hat{q} = 3.58$ for Adult and 3.71 for German credit), set membership, Pareto-membership probabilities $\hat{\pi}_m$, and the bootstrap frequency of being the best model by the equal-weight composite score.

On Adult, all nine models are in the TCS: the largest dominance statistic is 0.59 (naive Bayes), far below the critical value. Naive Bayes is clearly worst on error (0.406) and calibration (0.404), yet it is not significantly dominated: its equalised-odds gap (0.141) lies within the range of the other models' gaps (0.082–0.157) while the bootstrap standard deviation of a single gap is about 0.02, and dominance on all four criteria is required. The Pareto front of the estimates has eight members, and the Pareto-membership probabilities are 1.00 for seven models, 0.84 for the depth-12 tree and 0.27 for naive Bayes. The composite-score winner is not identified: the multilayer perceptron, 25-nearest neighbours and the depth-3 tree were best in 41%, 41% and

Table 3: Adult (sex), $n_{\text{test}} = 19{,}537$. S_m: dominance statistic (critical value 3.58). Naive: on the Pareto front of the point estimates. $\hat{\pi}_m$: bootstrap probability of Pareto-front membership. Top-1: bootstrap frequency of minimising the composite score.

Model	Error	EO gap	ECE	Noise drop	S_m	TCS	Naive	$\hat{\pi}_m$	Top-1
LR	0.150	0.110	0.007	0.031	-3.32	yes	yes	1.00	0.06
LR-strong-reg	0.152	0.157	0.015	0.023	-2.37	yes	yes	1.00	0.00
Tree(d=3)	0.155	0.093	0.004	0.042	-1.18	yes	yes	0.99	0.12
Tree(d=12)	0.146	0.082	0.034	0.065	-0.11	yes	yes	0.84	0.00
RandomForest	0.136	0.121	0.020	0.039	-6.63	yes	yes	1.00	0.00
HistGB	0.129	0.092	0.009	0.072	-5.03	yes	yes	1.00	0.00
kNN(25)	0.157	0.128	0.012	0.004	-7.44	yes	yes	1.00	0.41
NaiveBayes	0.406	0.141	0.404	0.023	0.59	yes	no	0.27	0.00
MLP	0.147	0.084	0.019	0.034	-2.04	yes	yes	1.00	0.41

Table 4: German credit (age), $n_{\text{test}} = 400$. Critical value 3.71.

Model	Error	EO gap	ECE	Noise drop	S_m	TCS	Naive	$\hat{\pi}_m$	Top-1
LR	0.263	0.076	0.054	0.015	-0.02	yes	yes	0.64	0.01
LR-strong-reg	0.285	0.013	0.103	-0.010	-0.67	yes	yes	0.80	0.06
Tree(d=3)	0.297	0.035	0.035	-0.005	-0.84	yes	yes	0.94	0.04
Tree(d=12)	0.310	0.056	0.287	-0.042	-0.87	yes	yes	0.87	0.01
RandomForest	0.247	0.053	0.054	-0.017	-1.41	yes	yes	0.99	0.57
HistGB	0.253	0.117	0.093	-0.020	-0.14	yes	yes	0.86	0.22
kNN(25)	0.278	0.101	0.041	-0.003	-0.70	yes	yes	0.78	0.08
NaiveBayes	0.297	0.092	0.229	0.008	0.51	yes	no	0.13	0.00
MLP	0.263	0.178	0.055	-0.010	0.03	yes	no	0.75	0.02

12% of resamples, and the 95% bootstrap intervals of the composite rank of the perceptron, kNN and the depth-3 tree all span ranks 1–5 (Figure 3). Under 5,000 random weight vectors drawn from a Dirichlet(1,1,1,1) distribution, the perceptron was best for 48% and kNN for 44%.

On German credit all nine models are again in the TCS, whereas the naive front excludes naive Bayes and the perceptron, the latter having a Pareto-membership probability of 0.75. The random forest is best by composite score in 57% of resamples, and its rank interval is 1–5. Every other model has an interval that reaches rank 7 or beyond. With only 400 test instances, and a smaller group among them, differences in the equalised-odds gap are not resolvable.

Fewer criteria. The TCS shrinks when fewer criteria are required (Table 5), because dominance on fewer coordinates is easier to certify. With error and calibration only, Adult yields a TCS of five models against three on the naive front; with error and the equalised-odds gap it yields seven against two. On German credit no pair of criteria allows any exclusion. The naive front is never larger than the TCS, which illustrates the central point: the point-estimate front looks more decisive than the data warrant.

6 Discussion

What the set says. A TCS of size M does not mean that all models are equally trustworthy. It means that, at the stated level, the test data do not contradict the claim that each is not

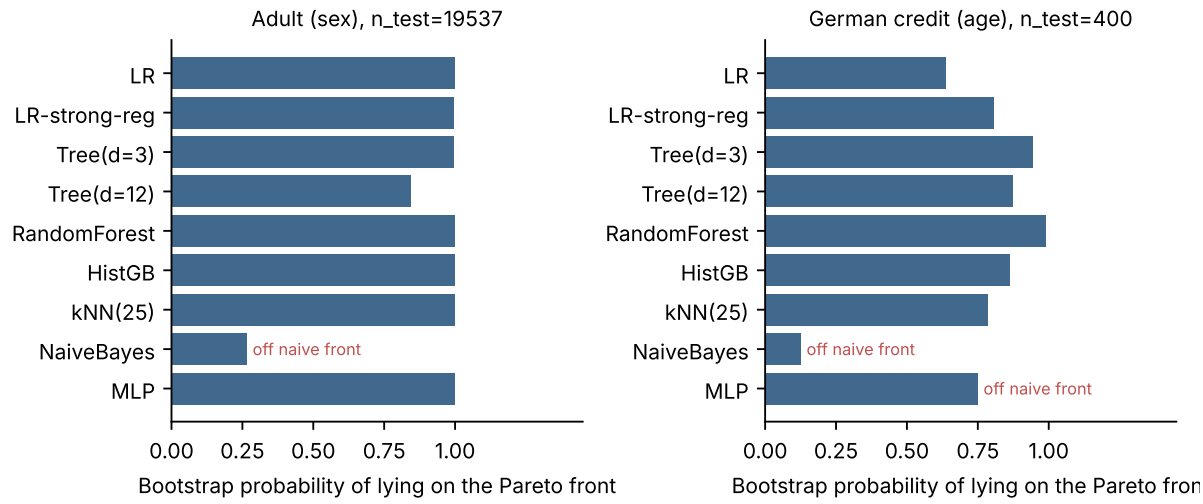


Figure 2: Bootstrap probability of Pareto-front membership. Labels mark models that are off the Pareto front of the point estimates.

Table 5: Number of models retained, for each subset of criteria: TCS and the Pareto front of the point estimates (EO: equalised-odds gap; noise: noise drop).

Criteria subset	Adult		German credit	
	TCS	Naive	TCS	Naive
err + EO	7	2	9	2
err + ECE	5	3	9	4
err + noise	7	6	9	3
EO + ECE	7	4	9	2
EO + noise	9	4	9	3
ECE + noise	7	3	9	4
err + EO + ECE	8	5	9	5
err + EO + noise	9	7	9	4
err + ECE + noise	7	7	9	6
EO + ECE + noise	9	6	9	5
err + EO + ECE + noise	9	8	9	7

strictly dominated. The practical consequences are that the choice among the retained models must rest on other grounds (cost, interpretability, domain constraints) and that, if a smaller set is wanted, one has to collect more test data, reduce the number of criteria or pre-specify a priority ordering.

Limitations. (i) The inference is conditional on the trained models and on one train/test split; retraining variance, seed variance and hyper-parameter search (Bouthillier et al., 2021; D'Amour et al., 2022) are not captured, and a refit-bootstrap or repeated-split extension would be needed to cover them. The case studies use a single split and a single perturbation draw. (ii) Proposition 1 is an asymptotic statement under bootstrap consistency; criteria such as ECE with binning and the equalised-odds gap with a maximum are not smooth, and the simulation uses criteria that are plain means. Our simulation therefore does not test the procedure on those non-smooth criteria. (iii) The procedure is conservative, as the simulation shows (retention above 99.7% against a nominal 95%) and costs power. (iv) The target is strict dominance; a

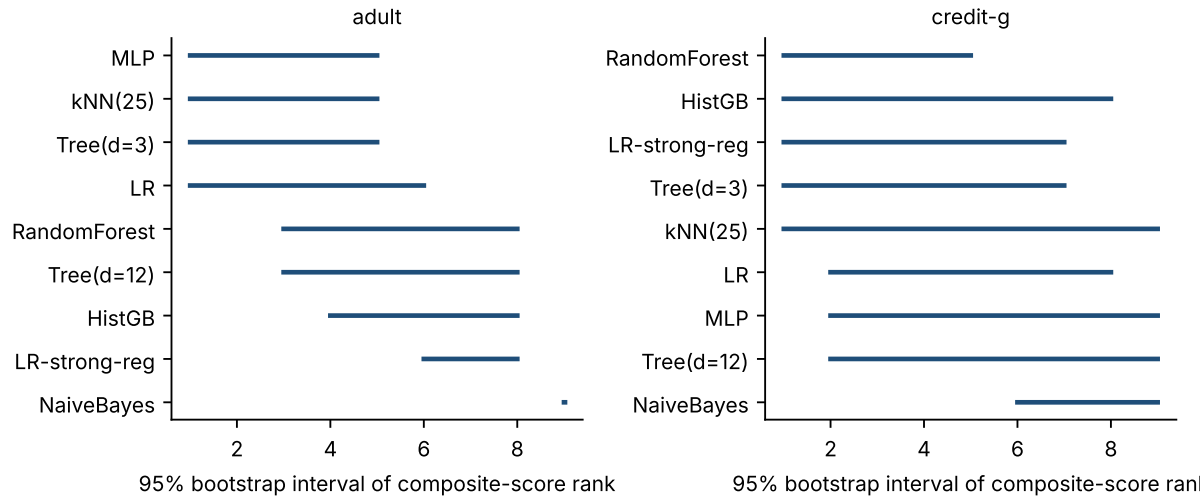


Figure 3: 95% bootstrap intervals for the rank of each model under the equal-weight composite score, models ordered by point-estimate rank.

model that is nearly dominated stays in $\mathcal{P}_0$. A practical-equivalence margin would be a useful refinement. (v) Candidate models are fixed in advance; selecting the list after looking at the test set invalidates the guarantee. (vi) We did not search exhaustively for prior work, so the claim of novelty is limited to the literature we reviewed.

Extensions. Natural directions are step-down refinements, a refit bootstrap that includes training variance, one-sided margins of practical equivalence, extension to explainability criteria such as attribution stability, and use of rank-based criteria from the SAFE family as coordinates.

Declarations

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Competing interests. The author declares no competing interests.

Data availability. Both data sets are public (OpenML: Adult, version 2; credit-g, version 1).

Code availability. All code, simulation outputs and figures are in the repository accompanying this submission (directories `tcs`, `experiments`, `results`).

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